DataBattle 2026

Feature Reference Manual & Evaluation Protocol

Complete mathematical definitions, sample values, and test mechanics

Team Documentation

March 2026

Contents

1 Overview of the Problem **2**
1.1 Goal . . . 2
1.2 Raw Data Structure . . . 2
1.3 Data Partitioning . . . 2
1.4 Segments . . . 3

2 How the Test Works **4**
2.1 What the evaluators give you . . . 4
2.2 What you submit . . . 4
2.3 Gain and Risk — the official evaluation protocol . . . 4
2.4 Is inference done strike-by-strike or as a batch? . . . 5

3 Sample Segment Used Throughout This Manual **6**

4 Feature Reference **7**
4.1 Group A — Amplitude Decomposition . . . 7
4.2 Group B — Segment Aggregations (full-segment statistics) . . . 7
4.3 Group C — Position Within Segment . . . 8
4.4 Group D — Temporal Dynamics (Lag Features) . . . 9
4.5 Group E — Cartesian Coordinates and Rolling Activity . . . 11
4.6 Group F — Silence Threshold Indicators . . . 12
4.7 Group G — Interaction Features . . . 12
4.8 Group K — Outer Ring Context (20–50 km) . . . 13
4.9 Group I — Calendar Features . . . 13
4.10 Group J — Airport Encoding . . . 14
4.11 Raw physical features passed directly . . . 14

5 Feature Validity Matrix **15**

6 The Leakage Problem: What the Model Actually Learned **16**
6.1 Observed performance . . . 16
6.2 Root cause . . . 16
6.3 The trade-off to decide on . . . 16

7 Sentinel Values for Single-Strike Segments **17**

8 Summary of All Features **17**

Overview of the Problem

Goal

For each cloud-to-ground (CG) lightning strike recorded inside a 20 km radius around one of five French airports, predict the probability that this strike is the **last CG strike** before a 30-minute silence period. That silence marks the official end of the alert and the point at which ground crews can safely resume operations.

$$\hat{y}_i = P(\text{strike } i \text{ is the last CG strike in its segment}) \in [0, 1]$$

Raw Data Structure

The raw dataset contains **507,071 rows**, one per detected lightning strike (both CG and IC). Each row provides:

Column	Type	Description
`lightning_id`	int	Global unique strike identifier
`airport`	string	Nearest airport (Ajaccio, Bastia, Biarritz, Nantes, Pise)
`airport_alert_id`	float/NaN	Per-airport alert counter. `NaN` if outside 20 km or IC strike
`date`	datetime	UTC timestamp of the strike
`amplitude`	float (kA)	Signed discharge current. Negative = normal CG return stroke
`maxis`	float	Peak current derivative
`icloud`	bool	`True` = cloud-to-cloud (IC), `False` = CG
`dist`	float (km)	Distance of strike from the airport
`azimuth`	float (°)	Compass bearing from airport to strike (0=N, 90=E)
`is_last_lightning_cloud_ground`	bool	**Target.** True = last CG before 30-min silence

Data Partitioning

Not all rows are used as model rows. Three populations are distinguished:

Population	Rows	Role
IC strikes (`icloud=True`)	378,079	Discarded entirely
Outside-zone CG (`dist > 20 km`)	72,393	Used only to compute outer-ring context feature
Inside-zone CG	**56,599**	**Model rows — the only rows scored**

Segments

A **segment** is one complete storm alert at one airport. It begins when a CG strike occurs inside 20 km and ends after 30 minutes of silence.

$$\texttt{segment_key} = \texttt{airport} \parallel \texttt{airport_alert_id} \quad \text{(e.g., “Ajaccio_1”, “Biarritz_47”)} \tag{1}$$

There are **2,744 segments** in training and **577 segments** in the test set. Within each segment, **exactly one strike** has `is_last = True`.

How the Test Works

What the evaluators give you

The test file (`dataset_set.csv`) contains **183,945 rows**, of which 13,780 are inside-zone CG strikes in 577 complete segments. The label column `is_last_lightning_cloud_ground` is set to `False` for every row — the answer is hidden.

Critical fact about test data

The test segments are **complete**: they include all CG strikes from alert start through to the true last strike. You are given the entire sequence. This is not a streaming problem.

What you submit

Your submission is a CSV with **one row per inside-zone CG strike**:

Column	Meaning
`airport`	Airport name
`airport_alert_id`	Alert counter (integer)
`prediction_date`	Timestamp of the strike being scored
`predicted_date_end_alert`	Your estimate of when the alert ends
`confidence`	Model output: $\hat{y}_i \in [0, 1]$

In our current setup, `predicted_date_end_alert = prediction_date` for every row (we predict the alert ends at the time of the strike itself).

Gain and Risk — the official evaluation protocol

For a threshold $\theta \in (0, 1)$, the evaluator processes each segment (a, k) as follows:

1. Select all rows of segment (a, k) where `confidence` $> \theta$.
2. Take the **earliest** `predicted_date_end_alert` among those rows — call it $\hat{t}_{a,k}$.
3. Compute the gain for this segment:

$$g_{a,k} = \underbrace{(t^*_{a,k} + 1800)}_{\text{baseline end}} - \hat{t}_{a,k} \quad \text{[seconds]}$$

where $t^*_{a,k}$ is the true last CG strike timestamp and +1800 seconds is the 30-minute operational baseline. If $g_{a,k} > 0$, you saved time. If $g_{a,k} < 0$, you ended the alert too early.

$$\text{Gain}(\theta) = \frac{1}{3600} \sum_{(a,k)} g_{a,k} \quad \text{[hours saved]}$$

$$\text{Risk}(\theta) = \frac{\#\{\text{dangerous strikes after } \hat{t}_{a,k} \text{ with dist} < 3\,\text{km}\}}{\#\{\text{all dangerous strikes}\}}$$

The best threshold θ^* is chosen as:

$$\theta^* = \arg\max_{\theta} \ \text{Gain}(\theta) \quad \text{subject to} \quad \text{Risk}(\theta) < 0.02$$

Is inference done strike-by-strike or as a batch?

Full segment batch. You receive all 13,780 test CG strikes at once, already grouped into complete segments. You compute all features over the full segment, then output one confidence score per strike. There is no streaming, no real-time state, no need to maintain history between strikes.

This is why features that use the complete segment (segment size, reverse rank, etc.) are technically computable at inference time in this competition.

Sample Segment Used Throughout This Manual

All feature examples below use **Ajaccio_1**: a 5-strike segment on January 2, 2016.

ID	Time (UTC)	Amplitude	Dist (km)	Azimuth (°)	is_last
4	21:22:53	−7.51 kA	14.79	20.85	False
5	21:24:46	−6.01 kA	15.12	29.06	False
6	21:25:59	−23.96 kA	15.58	18.93	False
7	21:27:04	−11.65 kA	15.34	38.64	False
9	21:28:54	−5.31 kA	17.56	38.77	**True**

Segment summary: $N = 5$ strikes, duration = 361 s = 6.02 min, mean distance = 15.68 km.

Feature Reference

Group A — Amplitude Decomposition

`amplitude`

Raw signed discharge current from the sensor. No transformation.

$$\texttt{amplitude}_i = a_i \in \mathbb{R} \quad [\text{kA}]$$

Physical meaning: negative values indicate the standard negative cloud-to-ground return stroke. Positive values are less common and tend to occur more frequently late in a storm's lifecycle as the charge structure becomes disorganised.

Ajaccio_1: $-7.51,\ -6.01,\ -23.96,\ -11.65,\ -5.31$ kA

`amp_magnitude`

Absolute intensity, removing the sign ambiguity.

$$\texttt{amp_magnitude}_i = |a_i|$$

Ajaccio_1: $7.51,\ 6.01,\ 23.96,\ 11.65,\ 5.31$ kA

`amp_is_positive`

Binary polarity flag.

$$\texttt{amp_is_positive}_i = \mathbf{1}[a_i > 0]$$

Physical meaning: the fraction of positive CG strikes increases as a storm decays. A rising positive ratio at the segment level (`seg_pos_cg_ratio`) is a known storm-ending indicator in the meteorological literature.

Ajaccio_1: $0, 0, 0, 0, 0$ (all negative — early-stage storm character)

Group B — Segment Aggregations (full-segment statistics)

Every strike in the segment receives the **same value** for these features. They describe the storm as a whole and require the complete segment to compute.

Let segment s contain N_s strikes with indices $\mathcal{S} = \{i : \texttt{segment_key}_i = s\}$.

`seg_size_cg`

$$\texttt{seg_size_cg}_i = N_s = |\mathcal{S}|$$

Ajaccio_1: all 5 strikes receive value 5.

`seg_mean_amp`

$$\texttt{seg_mean_amp}_i = \frac{1}{N_s} \sum_{j \in \mathcal{S}} a_j$$

Ajaccio_1: $\dfrac{-7.51 - 6.01 - 23.96 - 11.65 - 5.31}{5} = -10.888$ kA

`seg_mean_mag`

$$\texttt{seg_mean_mag}_i = \frac{1}{N_s} \sum_{j \in \mathcal{S}} |a_j|$$

Ajaccio_1: $\dfrac{7.51 + 6.01 + 23.96 + 11.65 + 5.31}{5} = 10.888$ kA

`seg_duration_min`

$$\texttt{seg_duration_min}_i = \frac{t_{\max(\mathcal{S})} - t_{\min(\mathcal{S})}}{60} \quad \text{[minutes]}$$

Ajaccio_1: (21:28:54 − 21:22:53) = 361 s = 6.017 min

`seg_mean_dist`

$$\texttt{seg_mean_dist}_i = \frac{1}{N_s} \sum_{j \in \mathcal{S}} d_j \quad \text{[km]}$$

Ajaccio_1: $\dfrac{14.79 + 15.12 + 15.58 + 15.34 + 17.56}{5} = 15.68$ km

`seg_pos_cg_ratio`

$$\texttt{seg_pos_cg_ratio}_i = \frac{1}{N_s} \sum_{j \in \mathcal{S}} \mathbf{1}[a_j > 0]$$

Ajaccio_1: 0/5 = 0.0 (no positive strikes in this storm)

Group C — Position Within Segment

Let r_i be the 0-indexed temporal rank of strike i within its segment ($r_i = 0$ for the first strike, $r_i = N_s - 1$ for the last).

`rank_in_seg`

Forward position, 0-indexed.

$$\texttt{rank_in_seg}_i = r_i \in \{0, 1, \ldots, N_s - 1\}$$

Ajaccio_1: $0, 1, 2, 3, 4$

`pct_position`

Fractional progress through the segment.

$$\texttt{pct_position}_i = \frac{r_i}{N_s}$$

Note the denominator is N_s (not $N_s - 1$), so the last strike receives $\frac{N_s - 1}{N_s} < 1$.

Ajaccio_1: 0.0, 0.2, 0.4, 0.6, **0.8** (last strike = 0.8, not 1.0)

`rank_rev_cg` — leakage candidate

Warning — Structural Leakage

This feature equals zero **if and only if** the strike is the last CG strike in the segment. Because the test data provides complete segments, this value is also 0 for the true last test strike. The model achieves AUC ≈ 1.0 primarily by learning this rule in 1–2 tree splits.

Reverse rank — counts from the end of the segment.

$$\texttt{rank_rev_cg}_i = (N_s - 1) - r_i = \begin{cases} 0 & \text{if } i \text{ is the last CG strike} \\ k & \text{if there are } k \text{ more strikes after } i \end{cases}$$

Ajaccio_1: $4, 3, 2, 1, \mathbf{0}$ (last strike = 0)

Measured correlation with target:

	`is_last = False`	`is_last = True`
`rank_rev_cg = 0`	59 (2.3%)	2,568 (97.7%)
`rank_rev_cg > 0`	53,913 (100%)	59 (0.1%)

Precision = 97.7%, Recall = 97.8%, AUC of this feature alone ≈ 0.998.

Group D — Temporal Dynamics (Lag Features)

Each feature compares strike i to the **previous strike** $i - 1$ in the same segment. The first strike of every segment has `NaN` for all Group D features (no predecessor exists). Single-strike segments receive sentinel values.

`time_since_prev`

Inter-strike gap in seconds.

$$\texttt{time_since_prev}_i = \begin{cases} t_i - t_{i-1} & \text{if } i > 0 \text{ in segment} \\ \texttt{NaN} & \text{if } i = 0 \text{ (first strike)} \\ 99999 & \text{if single-strike segment (sentinel)} \end{cases}$$

Physical meaning: a growing inter-strike gap signals the storm is decelerating. The 30-minute operational threshold means gaps of 600 s, 900 s, and 1,200 s are especially informative (see Group F).

Ajaccio_1:

Strike	Time	$t_i - t_{i-1}$ (s)	`time_since_prev`
4	21:22:53	—	NaN
5	21:24:46	113	113
6	21:25:59	73	73
7	21:27:04	65	65
9	21:28:54	110	110

`dist_delta`

Change in distance from airport — positive = storm moving away.

$$\texttt{dist_delta}_i = d_i - d_{i-1} \quad [\text{km}]$$

Ajaccio_1: NaN, +0.33, +0.46, −0.24, **+2.22**

The last strike moved +2.22 km further from the airport — consistent with storm exit.

`mag_delta`

Change in absolute amplitude — negative = storm weakening.

$$\texttt{mag_delta}_i = |a_i| - |a_{i-1}| \quad [\text{kA}]$$

Ajaccio_1: NaN, −1.50, +17.95, −12.31, **−6.34**

`storm_speed`

How far did the storm centroid move between consecutive strikes?

$$\texttt{storm_speed}_i = \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2} \quad [\text{km}]$$

where $\Delta x_i = x_i - x_{i-1}$, $\Delta y_i = y_i - y_{i-1}$ (Cartesian, see Group E).

Note: this is spatial displacement, not velocity (there is no division by time). A fast-moving storm exits the airport zone quickly, producing shorter alerts.

Group E — Cartesian Coordinates and Rolling Activity

`strike_x, strike_y`

Convert polar coordinates (distance, bearing) to a flat Cartesian map centred on the airport.

$$\texttt{strike_x}_i = d_i \cdot \sin(\alpha_i) \quad \text{[km East of airport]} \tag{2}$$
$$\texttt{strike_y}_i = d_i \cdot \cos(\alpha_i) \quad \text{[km North of airport]} \tag{3}$$

where α_i is azimuth in radians.

Ajaccio_1 last strike ($d = 17.56$ km, $\alpha = 38.77°$):

$$x = 17.56 \cdot \sin(38.77°) = 10.998 \text{ km East}, \quad y = 17.56 \cdot \cos(38.77°) = 13.692 \text{ km North}$$

`cg_count_10min, cg_count_30min`

CG activity rate in a backward time window, scoped to the segment.

$$\texttt{cg_count_W_min}_i = |\{j \in \mathcal{S} : t_i - W \cdot 60 < t_j \leq t_i\}|$$

Physical meaning: a falling count over successive strikes signals the storm is running out of energy. This is the most direct measure of storm deceleration.

Ajaccio_1 (all 5 strikes fall within a 6-min window, so both counts are identical): $1, 2, 3, 4, 5$

`rolling_mean_mag_10min`

Mean absolute amplitude over the last 10 minutes — is the storm weakening?

$$\texttt{rolling_mean_mag_10min}_i = \frac{\sum_{j \in \mathcal{S},\, t_j \in (t_i - 10', t_i]} |a_j|}{|\{j \in \mathcal{S} : t_j \in (t_i - 10', t_i]\}|}$$

Ajaccio_1: 7.51, 6.76, 12.49, 12.28, 10.89

`rolling_pos_ratio_10min`

Fraction of positive CG strikes in the last 10 minutes — rising fraction = decay phase.

$$\texttt{rolling_pos_ratio_10min}_i = \frac{\sum_{j \in \mathcal{S},\, t_j \in (t_i - 10', t_i]} \mathbf{1}[a_j > 0]}{|\{j \in \mathcal{S} : t_j \in (t_i - 10', t_i]\}|}$$

`rolling_mean_dist_10min`

Mean distance over the last 10 minutes — rising value = storm drifting away.

$$\texttt{rolling_mean_dist_10min}_i = \frac{\sum_{j \in \mathcal{S},\, t_j \in (t_i - 10', t_i]} d_j}{|\{j \in \mathcal{S} : t_j \in (t_i - 10', t_i]\}|}$$

Group F — Silence Threshold Indicators

These are binary flags that encode whether the gap **before** this strike exceeded an operationally significant threshold. They look **backward** — they flag silence that already happened before this strike, not silence that will follow it.

$$\texttt{silence_over_10min}_i = \mathbf{1}[\texttt{time_since_prev}_i > 600] \tag{4}$$
$$\texttt{silence_over_15min}_i = \mathbf{1}[\texttt{time_since_prev}_i > 900] \tag{5}$$
$$\texttt{silence_over_20min}_i = \mathbf{1}[\texttt{time_since_prev}_i > 1200] \tag{6}$$

Physical meaning: a gap of over 20 minutes before this strike means the storm nearly ended, then produced one more strike. The operational threshold is 30 minutes, so 20-minute gaps are very late in the storm lifecycle.

Ajaccio_1: all zeros — longest gap is 113 s, well below any threshold.

Measured signal (training set, 56,599 rows):

Feature = 1	`is_last=False`	`is_last=True`
`silence_over_10min`	4.0%	26.7%
`silence_over_15min`	4.3%	30.0%
`silence_over_20min`	4.4%	30.4%

These features are informative but not leaky — many non-last strikes also follow long gaps (mid-segment pauses).

Group G — Interaction Features

`silence_x_dist_away`

Combines a long silence **and** storm moving away — the strongest composite end signal.

$$\texttt{silence_x_dist_away}_i = \texttt{silence_over_15min}_i \times \mathbf{1}[\texttt{dist_delta}_i > 0]$$

This equals 1 only when both conditions hold simultaneously: the gap before this strike exceeded 15 minutes *and* the storm moved further from the airport. This joint condition is very specific to end-of-storm behaviour.

`weak_and_moving_away`

Combines below-average amplitude **and** increasing distance.

$$\texttt{weak_and_moving_away}_i = \mathbf{1}[|a_i| < \texttt{seg_mean_mag}_i] \times \mathbf{1}[\texttt{dist_delta}_i > 0]$$

Ajaccio_1 last strike: $|a_9| = 5.31 < \texttt{seg_mean_mag} = 10.888$ ✓ and $\texttt{dist_delta}_9 = +2.22 > 0$ ✓ $\Rightarrow$ value $= 1$

Group K — Outer Ring Context (20–50 km)

`outer_ring_cg_10min`

CG activity in the broader 20–50 km zone around the airport in the last 10 minutes. Uses outside-zone strikes (`airport_alert_id = NaN`) which are **not** model rows but are used here as context.

$$\texttt{outer_ring_cg_10min}_i = |\{j \in \mathcal{O}_a : \; t_i - 600 < t_j \leq t_i\}|$$

where $\mathcal{O}_a$ = set of outside-zone CG strikes for airport a (dist > 20 km).

Computed via `pandas.merge_asof` (backward join, tolerance 10 min) for efficiency.

Physical meaning: if the outer ring is still active, the storm cell has not exhausted its energy supply and inner-zone CG strikes are likely to continue. If the outer ring goes quiet simultaneously with the inner zone, the whole system is collapsing.

Measured signal (EDA, training set):

$$\text{Mean outer ring count:} \quad \underbrace{22.5}_{\text{non-last strikes}} \quad \text{vs} \quad \underbrace{2.4}_{\text{last strikes}}$$

A **10x difference** — without using any future information. This is the strongest single physically-grounded feature.

Group I — Calendar Features

$$\texttt{hour}_i = \text{UTC hour of } t_i \quad \in \{0, \ldots, 23\} \tag{7}$$

$$\texttt{month}_i = \text{calendar month of } t_i \quad \in \{1, \ldots, 12\} \tag{8}$$

$$\texttt{is_summer}_i = \mathbf{1}[\texttt{month}_i \in \{6, 7, 8\}] \tag{9}$$

$$\texttt{is_afternoon}_i = \mathbf{1}[\texttt{hour}_i \in \{12, \ldots, 18\}] \tag{10}$$

EDA findings:

- All five airports peak 12–17 UTC (afternoon convective storms driven by solar heating).
- **Mediterranean regime** (Ajaccio, Bastia, Pise): peak activity June–August.

- **Atlantic/Po Valley regime** (Biarritz, Nantes): peak activity May.
- Summer afternoon storms tend to be more intense and longer than winter events.

Group J — Airport Encoding

`airport_target_enc`

Per-airport empirical probability that a CG strike is the last in its segment.

$$\texttt{airport_target_enc}_i = \frac{\sum_{j:\texttt{airport}_j=a} \mathbf{1}[y_j = 1]}{\sum_{j:\texttt{airport}_j=a} 1} \quad \text{where } a = \texttt{airport}_i$$

Computed from the **training fold only** in cross-validation — never from the validation fold. This prevents the feature from leaking validation-set labels. For inference, it is computed from the full training set.

Example: Ajaccio has ≈ 0.0498 (roughly 1 last strike per 20 CG rows).

Raw physical features passed directly

These raw columns are passed to LightGBM unchanged to allow the model to discover non-linear threshold effects on its own:

Feature	Meaning
`dist`	Distance of this strike from airport (km)
`azimuth`	Compass bearing from airport to strike (°)
`maxis`	Peak current derivative (sensor-specific signal quality indicator)
`strike_x`, `strike_y`	Cartesian position relative to airport (km East, km North)

Feature Validity Matrix

The central question is: *can a feature be legitimately computed on the test data, and does it introduce leakage of the target?*

Feature(s)	Uses future data?	Computable on test?	Notes
`amplitude`, `amp_magnitude`, `amp_is_positive`	No	Yes	Current strike only
`seg_size_cg`, `seg_mean_*`, `seg_duration_min`, `seg_mean_dist`, `seg_pos_cg_ratio`	Yes — full segment	Yes — test gives complete segments	Storm-level aggregate
`rank_in_seg`, `pct_position`	Yes — full segment	Yes	Position, not tautological
`rank_rev_cg`	**Yes**	**Yes — equals 0 for the true last strike**	**Structural leakage:** value 0 ≡ is_last=True
`time_since_prev`, `dist_delta`, `mag_delta`, `storm_speed`	No — looks backward	Yes	Previous-strike delta
`cg_count_*`, `rolling_mean_*`, `rolling_pos_ratio_*`, `rolling_mean_dist_*`	No — backward window	Yes	Backward rolling window
`silence_over_10/15/20min`	No — gap before strike	Yes	Not tautological (many false positives)
`silence_x_dist_away`, `weak_and_moving_away`	Partially	Yes	Interaction of backward features
`outer_ring_cg_10min`	No — backward 10-min	Yes	External context
`hour`, `month`, `is_summer`, `is_afternoon`	No	Yes	Calendar
`airport_target_enc`	No (CV-safe)	Yes — from training set	Encoding
`dist`, `azimuth`, `maxis`, `strike_x`, `strike_y`	No	Yes	Raw physical

The Leakage Problem: What the Model Actually Learned

Observed performance

CV Fold	AUC	F1	Brier
Fold 1	1.00000	0.9953	0.000265
Fold 2	0.99999	0.9934	0.000532
Fold 3	0.99999	0.9943	0.000454
Fold 4	0.99999	0.9906	0.000649
Fold 5	1.00000	0.9962	0.000293
Temporal stress test (train $\leq$ 2020, val $\geq$ 2021)	1.00000	0.9983	0.000254

Root cause

The model learned a two-split rule equivalent to:

```
if rank_rev_cg == 0:
    predict confidence = 0.9999 # last strike
else:
    predict confidence = 0.0005 # not last
```

This is not overfitting (scores are consistent across all folds and across the temporal split). It is **structural leakage**: the feature `rank_rev_cg` is mathematically equivalent to the target when computed on complete segments.

OOF prediction distribution:

$$\overline{\hat{y}}_{\text{is_last=True}} = 0.9991, \quad \overline{\hat{y}}_{\text{is_last=False}} = 0.0005$$

The trade-off to decide on

	Keep `rank_rev_cg`	**Remove** `rank_rev_cg`
AUC	≈ 1.00	≈ 0.83–0.90
Competition gain	30 min/alert (maximum)	Potentially $>$ 30 min if model predicts end earlier
Competition risk	Near 0%	Depends on threshold
What the model learns	"Count from end of list"	Storm physics, spatial decay, temporal gaps
SHAP explainability	Trivial	Meaningful
Evaluator impression	Suspicious (AUC=1.0)	Credible

Sentinel Values for Single-Strike Segments

A single-strike segment (exactly one CG strike, immediately followed by 30 min of silence) has `NaN` for all Group D and E lag/rolling features because no previous strike exists. These are filled with sentinels that encode "first and only strike" semantics:

Feature	Sentinel value	Interpretation
`time_since_prev`	99999	Very long prior silence (unknown origin)
`dist_delta`	0	No movement information
`mag_delta`	0	No amplitude change information
`storm_speed`	0	No movement information
`cg_count_10min`	1	Only 1 known strike
`cg_count_30min`	1	Only 1 known strike
`rolling_pos_ratio_10min`	0	No prior ratio
`rolling_mean_mag_10min`	dataset median	Neutral (data-driven)
`rolling_mean_dist_10min`	dataset median	Neutral (data-driven)
`silence_over_10/15/20min`	0	No prior silence observed

Summary of All Features

Feature	Group	Formula / Definition
`amplitude`	A	Raw signed discharge current a_i (kA)
`amp_magnitude`	A	$\lvert a_i \rvert$
`amp_is_positive`	A	$\mathbf{1}[a_i > 0]$
`seg_size_cg`	B	N_s = total CG strikes in segment
`seg_mean_amp`	B	$\frac{1}{N_s}\sum_{j\in\mathcal{S}} a_j$
`seg_mean_mag`	B	$\frac{1}{N_s}\sum_{j\in\mathcal{S}} \lvert a_j \rvert$
`seg_duration_min`	B	$(t_{\max} - t_{\min})/60$ [min]
`seg_mean_dist`	B	$\frac{1}{N_s}\sum_{j\in\mathcal{S}} d_j$ [km]
`seg_pos_cg_ratio`	B	$\frac{1}{N_s}\sum_{j\in\mathcal{S}} \mathbf{1}[a_j > 0]$
`rank_in_seg`	C	r_i (0-indexed forward rank in segment)
`pct_position`	C	r_i/N_s
`rank_rev_cg`	C	$(N_s - 1) - r_i$ (0 = last strike)
`time_since_prev`	D	$t_i - t_{i-1}$ [s]
`dist_delta`	D	$d_i - d_{i-1}$ [km]
`mag_delta`	D	$\lvert a_i \rvert - \lvert a_{i-1} \rvert$ [kA]
`storm_speed`	D/E	$\sqrt{(\Delta x_i)^2 + (\Delta y_i)^2}$ [km]
`cg_count_10min`	E	$\lvert\{j \in \mathcal{S} : t_i - 600 < t_j \le t_i\}\rvert$
`cg_count_30min`	E	$\lvert\{j \in \mathcal{S} : t_i - 1800 < t_j \le t_i\}\rvert$
`rolling_mean_mag_10min`	E	Mean $\lvert a_j \rvert$ over last 10 min in segment
`rolling_pos_ratio_10min`	E	Mean $\mathbf{1}[a_j > 0]$ over last 10 min
`rolling_mean_dist_10min`	E	Mean d_j over last 10 min in segment

`silence_over_10min`	F	$\mathbf{1}[\texttt{time_since_prev}_i > 600]$
`silence_over_15min`	F	$\mathbf{1}[\texttt{time_since_prev}_i > 900]$
`silence_over_20min`	F	$\mathbf{1}[\texttt{time_since_prev}_i > 1200]$
`silence_x_dist_away`	G	$\texttt{silence_over_15min}_i \times \mathbf{1}[\texttt{dist_delta}_i > 0]$
`weak_and_moving_away`	G	$\mathbf{1}[\lvert a_i\rvert < \overline{\lvert a\rvert}_s] \times \mathbf{1}[\texttt{dist_delta}_i > 0]$
`outer_ring_cg_10min`	K	CG count in 20–50 km zone, last 10 min
`hour`	I	UTC hour $\in \{0, \ldots, 23\}$
`month`	I	Calendar month $\in \{1, \ldots, 12\}$
`is_summer`	I	$\mathbf{1}[\text{month} \in \{6, 7, 8\}]$
`is_afternoon`	I	$\mathbf{1}[\text{hour} \in \{12, \ldots, 18\}]$
`maxis`	raw	Peak current derivative
`dist`	raw	Distance from airport (km)
`azimuth`	raw	Bearing from airport (°)
`strike_x`	raw	$d_i \sin(\alpha_i)$ (km East)
`strike_y`	raw	$d_i \cos(\alpha_i)$ (km North)
`airport_target_enc`	J	Per-airport mean(is_last) from training fold

Table 2: Complete feature list (37 features passed to LightGBM)